

Analysis of Scientific Data



Lecture 9

Model Extensions

Research Project

- Study design involve **one response variable and at least two explanatory variables**
 - ▶ two-way ANOVA
 - ▶ multiple linear regression
- Minimum sample size:
 - ▶ experimental studies: 24
 - ▶ observational studies: 48
- Can work individually or in a group of two or three students:
 - ▶ Group registration due at **4pm on Friday, 15 May 2026**.
 - ▶ Once you join a group, **you will not be able to leave it**
- Use the powerpoint template on Blackboard

Recall the maternal immunisation study from the Week 7 lecture:

- Is there a difference in mean serum concentrations (mg/dL) of pertussis-specific IgG antibodies between the three breeds? ✓
- Does the vaccine have an effect on serum concentrations (mg/dL) of pertussis-specific IgG antibodies?
- Is the vaccine effect the same for all breeds?

Recap: One-way ANOVA

Is there a difference in mean serum concentrations (mg/dL) of pertussis-specific IgG antibodies between the three breeds?

Test $H_0 : \mu_H = \mu_L = \mu_B$ vs H_1 : not all means are equal

Source	DF	SS	MS	F	p-value
Breed	2	1000819	500409	1.909	0.172
Residuals	21	5503632	262077.9		
Total	23	6504451			

$$R^2 = 1 - \frac{SSR}{SST} = \frac{1000819}{6504451} = 0.154$$

Breed explains about 15.4% of the variability.

Poll Question

What are the **group** degrees of freedom for the one-way ANOVA of IgG concentration by breed?

1. 1

2. 2

3. 3

4. 4

Poll Question

What are the **residual** degrees of freedom for the one-way ANOVA of IgG concentration by breed?

1. 18
2. 19
3. 20
4. 21
5. 22
6. 23
7. 24

Poll Question

What are the **residual** degrees of freedom for the one-way ANOVA of IgG concentration by vaccine?

1. 18

2. 19

3. 20

4. 21

5. 22 ✓

6. 23 ✗

7. 24

$$n = 24$$

$$n - \text{number of groups} = n - 2 = 22$$

$$R^2 = \frac{2520072}{2520072 + 3984378} = 0.3874$$

$$MSR = 181108 = S_p^2 \quad \leftarrow \text{pooled variance}$$

Two-way ANOVA

Is the vaccine effect the same for all breeds?

How many residual degrees of freedom?

$$n - \text{number of groups} = 24 - 6 = 18$$

What's the residual sum of square?

$$MS = \frac{SS}{DF}$$

$$P(F_{1,18} \geq 15.38)$$

Source	DF	SS	MS	F	P
Vaccine	1	2520072	2520072	15.38	0.000449
Breed	2	1000818	500409	3.05	0.0723
Interaction	2	34504	17252.5	0.105	0.9009
Residual	18	2949056	163836.4		
Total	23	6504450			

Two-way ANOVA

Final Examination, Semester 2, 2017 – Question 2

In a study of factors thought to be associated with birth weight, a random sample of 100 births was selected from all the birth records at a hospital in 2001. Three variables were extracted from each record:

- length of gestation (weeks)
- smoking status of the mother (smoker/nonsmoker)
- birth weight (grams)

Is there any evidence that smoking has an effect on birth weight?

Poll Question

(a) The mean length of gestation for the 100 births was 38.36 weeks with standard deviation 3.36 weeks. Construct a 95% confidence interval for the mean length of gestation for all births at this hospital in 2001.

What degrees of freedom do we use for calculating this confidence interval?

$$n = 100, \bar{x} = 38.36, s = 3.36$$

$$95\% \text{ CI is } \bar{x} \pm t_{df}^* (0.975) \times se(\bar{x})$$

$$= 38.36 \pm t_{99}^* (0.975) \times \frac{3.36}{\sqrt{100}}$$

$$= 38.36 \pm 1.984217 \times \frac{3.36}{\sqrt{100}}$$

$$= [37.69, 39.03]$$

1. 96

2. 97

3. 98

4. 99

5. 100

40 weeks?

(b) There are claims that the mean length of gestation has been decreasing in recent decades from the traditionally held value of 40 weeks. Based on the confidence interval, or otherwise, do the data from this hospital support or contradict such claims?

Birth Weight and Smoking Status

The main question of interest is whether babies born to smoking mothers tend to have lower birth weights.

(c) Compare the distributions of birth weights for smokers and non-smokers.

(d) On average, how much lower are birth weights for smokers compared to non-smokers?

(e) Carry out an appropriate t -test and summarise the conclusions.

$$H_0: \mu_S = \mu_N \quad \text{vs} \quad H_1: \mu_S < \mu_N$$

$$p = 0.09083$$

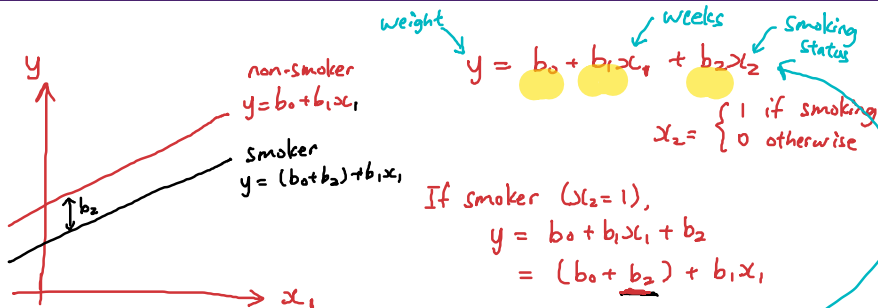
Weak evidence

Birth Weight and Gestation

There is considerable variability in birth weights for both smoking status groups. However, some of this variability may be explained by the different gestation lengths for each birth.

We can use **multiple regression** to model the effect of smoking on birth weight, *taking into account the relationship with gestation length*.

Multiple linear regression



If smoker ($x_2 = 1$),

$$y = b_0 + b_1 x_1 + b_2$$
$$= (b_0 + \underline{b_2}) + b_1 x_1$$

If non-smoker ($x_2 = 0$),

$$y = b_0 + b_1 x_1$$

Test $H_0: \beta_2 = 0$ vs $H_1: \beta_2 < 0$

$$t = \frac{b_2 - 0}{\text{se}(b_2)} =$$

P-value = $P(T \leq \quad)$

$+ b_3 \underline{x_1 x_2}$
interaction

Poll Question

What are the residual degrees of freedom for this multiple regression?

$$n = 3 = 97$$

estimate $\beta_0, \beta_1, \beta_2$

1. 96

2. 97 ✓

3. 98 ✗

4. 99

5. 100

Based on the model output:

1. What does the value '130.05' tell you?
2. What does the value '-294.40' tell you?
3. What does the value '-1724.42' tell you?
4. What is the estimated birth weight for a birth at 35 weeks gestation to a non-smoking mother?

Birth Weight and Smoking Status

- (f) Based on the multiple regression model, is there any evidence of a difference in mean birth weight between smoking and non-smoking mothers?
- (g) Briefly explain why the conclusion from the multiple regression model might be different to the conclusion from the two-sample t -test.

Cognitive Failures

A broad epidemiological study included the following question to measure cognitive failures experienced by an individual: 'Have you experienced problems of memory (e.g. forgetting where you put things), attention (e.g. failures of concentration) or action (doing the wrong thing)?' Responses were combined into a single binary response for each person – “yes” (experienced some type of cognitive failure) or “no” (experienced no cognitive failure).

One group of researchers using the data from this study were interested in whether caffeine consumption (mg/day) had an effect on cognitive failure.

Data based on Andrew P Smith (2008) Caffeine, cognitive failures and health in a non-working community sample. *Human Psychopharmacology* 24, 29-34.

Poll Question

What are the log odds of getting heads on a single toss of a fair coin?

$$\ln \left(\frac{p}{1-p} \right)$$

Odds

1. -1
2. 0
3. 0.5
4. 1

Poll Question

For the logistic regression model

The diagram shows the logistic regression model equation: $\ln\left(\frac{p}{1-p}\right) = b_0 + b_1x$. The left side, $\ln\left(\frac{p}{1-p}\right)$, is enclosed in a red box, and an arrow points from the word "odds" to this box. The right side, $b_0 + b_1x$, has an arrow pointing from the value "0.1464" to b_0 , and another arrow pointing from the value "-0.0039" to b_1 . The variable x is circled in red, and a red line is drawn underneath the entire equation.

$$\ln\left(\frac{p}{1-p}\right) = b_0 + b_1x$$

what value of x gives an even chance of the response?

1. b_1/b_0
2. $-b_1/b_0$
3. b_0/b_1
4. $-b_0/b_1$
- $$p = \frac{\text{odds}}{1 + \text{odds}}$$

Logistic regression